

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Nitrate + Nitrite	WQP 014	1.6	

PREPARED BY			
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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC
1.1	9/8/23	Increased sample volume and changed to acid addition in the lab according to QAPP.	TAC
1.2	11/30/23	Sample preservation changes, acid and base volumes. SOP instructions revised according to chemical reagent changes in pH. Addition of spike and CRM preparation instructions updated to improve detection.	TAC
1.3	6/10/24	Modification of CRM and Spike preparation instructions on Section 13 to include combined ERA Waters CRM. Addition of Spike preparation instructions to include a total volume of 10mL sample/blank. The instrument preparation section was moved before running the instrument start-up for practicality. Addition of instructions to retrieve report as ASCII file. Revision of the acidification and neutralization instructions.	TAC
1.4	9/4/24	Sample neutralization volume changed to 500 uL of NaOH for the 100 mL sample for more consistent results. The option to neutralize the 15 mL sample was eliminated.	TAC
1.5	5/27/25	Minor updates to improve method clarity	TAC

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1.6	6/6/25	Additional minor updates to improve method clarity	TAC
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1. Title

Standard Operating Procedure for the Determination of Nitrate and Nitrite in Seawater.

2. Applicability

This SOP applies to the determination of seawater Nitrate and Nitrite ($\text{NO}_3^-/\text{NO}_2^-$) at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater bottom $\text{NO}_3^-/\text{NO}_2^-$ at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater $\text{NO}_3^-/\text{NO}_2^-$ data following the Quality Assurance Project Plan.

4. Method

- 4.1. Standard Method EPA Method 353.2 (Environmental Protection Agency)

5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. Read each reagent SDS prior to use for safe handling and disposal.
- 5.3. Prepare reagents to be used on this SOP inside a hood.
- 5.4. Wear disposable latex gloves, lab coat, and safety glasses while performing this procedure.
- 5.5. Be aware of waste container liquid level to avoid spills.

6. General Precautions

- 6.1 Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2 Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.
- 6.3 Strictly follow SOPs and manufacturer instructions.
- 6.4 The photometer LED lamp must be changed with the instrument turned OFF.
- 6.5 Keep all instrument covers in place during analysis.
- 6.6 Ensure the supply of reagents and wash is adequate to complete the run.
- 6.7 Never attempt to turn off or change pump speed during a run, unless there is an emergency.
- 6.8 Never leave the pump platen engaged without pumping for long periods of time.
- 6.9 Ensure the waste tube is never submerged in the waste.

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7. Training

- 7.1 Personnel shall be trained in the proper sample collection, handling, and analysis procedure for $\text{NO}_3^-/\text{NO}_2^-$.
- 7.2 Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1 Field Sampling Log Form (Appendix F)
- 8.2 Consumables
 - 8.2.1 Kimwipes
 - 8.2.2 Latex Gloves
 - 8.2.3 Beaker (1000 mL) for solution preparation
 - 8.2.4 Graduated cylinder (1000 mL, 200 mL)
 - 8.2.5 Plastic/Glass bottles for reagent and wash storage (1000 mL)
 - 8.2.6 Plastic bottles for stock standard and cadmium column solutions storage (100 mL)
 - 8.2.7 Amber glass bottle for color reagent storage (1000 mL)
 - 8.2.8 125 mL polypropylene plastic bottle
 - 8.2.9 Plastic Syringe for Filtering Sample
 - 8.2.10 PES Filters (0.45 μm) 1 per seawater sample
- 8.3 SEAL AA500 AutoAnalyzer
- 8.4 Required Chemicals

Compound	CAS No.	Safety classification	
Ammonium chloride, NH_4Cl	12125-02-9	GHS07	Irritant
Ammonia, 25% solution, NH_3	1336-21-6	GHS04, GHS06, GHS08, GHS09	Warning-Compressed gas, Danger-Health hazard, Warning/Danger- Environmentally Damaging
Cadmium, particle size 0.3-0.8 mm	7440-43-9	GHS02, GHS06, GHS08, GHS09	Danger, Danger-toxic, Danger-Health hazard, Warning/Danger- Environmentally Damaging
Copper (II) sulfate pentahydrate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	7758-99-8	GHS07, GHS09	Irritant- Warning/Danger- Environmentally Damaging
Hydrochloric acid, 36.5-38%, HCl	7647-01-1	GHS05, GHS07	Warning/Danger – Corrosive, Irritant
N-(1-Naphthyl)ethylenediamine dihydrochloride, $\text{C}_{12}\text{H}_{14}\text{N}_2 \cdot 2\text{HCl}$	1465-25-4	GHS07	Irritant

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Compound (Cont.)	CAS No.	Safety classification	
Phosphoric acid, conc., H_3PO_4	7664-38-2	GHS05,	Warning/Danger – Corrosive
Potassium nitrate, KNO_3	7757-79-1	GHS03	Danger - Oxidizing
Sodium chloride, $NaCl$	7647-14-5	---	
Sodium hydrogen carbonate, $NaHCO_3$	144-55-8		
Sodium hydroxide	1310-73-2	GHS05	
Sodium hypochlorite 10% solution	7681-52-9	GHS05, GHS09	
Sodium nitrite, $NaNO_2$	7632-00-0	GHS03, GHS06, GHS09	Danger - Oxidizing, Danger – Toxic, Warning/Danger- Environmentally Damaging
Sulfanilamide, $4-NH_2C_6H_4SO_2NH_2$	63-74-1	---	

8.5 Reagents, Standards, and Preparation Instructions

8.5.1 Synthetic Seawater

Reagent	Amount	Instructions	
Sodium chloride	35 g	1. Weigh sodium chloride. 2. Add the measured amount into a beaker with a stir rod. 3. Add approximately 500-600 mL of distilled water to the beaker. 4. Stir on a plate until all the salt has dissolved. 5. Pour beaker contents into a 1L graduated cylinder. 6. Bring to volume with distilled water.	
Distilled water	1L		
General Details			
Storage	Plastic container at room temperature.		
Stability	One week.		
Purpose	Stock standard solutions preparation.		

8.5.2 Hydrochloric Acid, 1N

Reagent	Amount	Instructions
Hydrochloric acid 36.5-38%	85 mL	1. Add approximately 700 mL of distilled water into a beaker. 2. Measure HCl in a 100 mL graduated cylinder, slowly add it to step #1, and gently stir. 3. Let it cool to room temperature. 4. Pour beaker contents into a 1L graduated cylinder. 5. Bring to volume with distilled water.
Distilled water	1 L	
General Details		
Storage	Glass container at room temperature.	

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Stability	Indefinite. Replace if turbid.
Purpose	For weekly cleanup.

8.5.3 Sodium hydroxide (1 mol/L)

SIS Sodium Hydroxide (1 M, 1 L)		
Reagent	Amount	Instructions
Sodium hydroxide	40 g	1. Weigh the NaOH. 2. Add the weighed amount to a beaker. 3. Add approximately 500mL of distilled water to the beaker. 4. Place the beaker on a stir plate and mix. 5. Once well mixed, transfer contents to a graduated cylinder and bring to volume with distilled water.
Distilled water	1 L	
General details		
Storage	Glass container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	For weekly cleanup.	

8.5.4 Special wash solution

Disinfectant Solution		
Reagent	Amount	Instructions
Clorox bleach (Sodium hypochlorite 10% solution)	10 mL	1. Dilute 10 mL of Clorox bleach solution into 50 mL of distilled water.
Distilled water	50mL	
General Details		
Storage	Plastic container at room temperature.	
Stability	One day.	
Purpose	For weekly cleanup.	

8.5.5 Ammonium Chloride Reagent

Reagent	Amount	Instructions
Ammonium chloride 12125-02-9	20 g	1. Weigh the Ammonium chloride and transfer it to a beaker. 2. Add approximately 90 mL of distilled water into the beaker 3. Place the beaker on a stir plate and stir. 4. Adjust the pH to 7.5± 0.1 with the ammonia solution. 5. Transfer the solution to a graduated cylinder. 6. Bring to volume with distilled water. 7. Transfer to a storage container and mix thoroughly.
Ammonia Solution 25% 1336-21-6	pH adjust	
Distilled water	500 mL	
General Details		
Storage	Glass container at room temperature.	

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Stability	One week.
Purpose	For NOx Analysis & Cadmium Column Preparation.

8.5.6 Color Reagent

Reagent	Amount	Instructions
Phosphoric acid concentrate	50 mL	1. Weigh the N-1-naphthylethylenediamine dihydrochloride and add it to a beaker. 2. Weigh the sulfanilamide and add it to the beaker. 3. Add approximately 200 mL of distilled water to the beaker. 4. Measure the concentrated phosphoric acid in a graduated cylinder. 5. Place the beaker on a stir plate and dissolve the contents. 6. Slowly add the measured acid to the beaker. 7. Transfer contents to a graduated cylinder. 8. Bring to volume (500 mL) with distilled water and mix thoroughly.
Sulfanilamide 63-74-1	5 g	
N-1-naphthylethylenediamine dihydrochloride 1465-25-4	0.25g	
Distilled water	500 mL	
General Details		
Storage	Amber glass container at 4°C.	
Stability	One month.	
Purpose	For NOx Analysis. Note: The reagent should be colorless: if it is pink the phosphoric acid is probably impure.	

8.5.7 6N Hydrochloric Acid

Solution of Hydrochloric Acid		
Reagent	Amount	Instructions
Hydrochloric acid 36.5-38%	49.5 mL	1. Measure the Hydrochloric acid in a graduated cylinder. 2. Add approximately 40 mL of distilled water to a 100 mL graduated cylinder. 3. Slowly add the HCl to the graduated cylinder with distilled water. 4. Cool down to room temperature. 5. Bring to volume with distilled water.
Distilled water	100 mL	
General Details		
Storage	Glass or plastic container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	For cadmium column preparation.	

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8.5.8 Copper Sulfate solution 2%

Reagent	Amount	Instructions
Copper sulfate pentahydrate 7758-99-8	2 g	1. Weigh 2 g of copper sulfate. 2. Dilute to 100 mL with distilled water and mix thoroughly.
Distilled water	100 mL	
General Details		
Storage	Glass or plastic container at room temperature.	
Stability	Indefinite. Replace if turbid.	
Purpose	For cadmium column preparation.	

8.5.9 Stock standard A - Nitrate (NO₃), 1000 mg/L

Reagent	Amount	Instructions
Potassium Nitrate (KNO ₃) 7757-79-1	3.61g	1. Weigh Potassium Nitrate (KNO₃). 2. Transfer the KNO₃ to a 1000 mL beaker. 3. Add approximately 600 ml of distilled water. 4. Mix the contents on a stir plate. 5. Transfer contents to a graduated cylinder. 6. Bring to volume (1L) with synthetic seawater.
Distilled Water	500 mL	
General Details		
Storage	Glass or plastic container at 4°C.	
Stability	One year.	
Purpose	NO ₃ stock standard for calibration.	

8.5.10 Stock standard C - Nitrate (NO₃) 400 µmol/L

Reagent	Amount	Instructions
Stock Standard A 1000 mg/L NO ₃ -N	0.56 mL	1. Pipette 0.56 mL of stock standard A. 2. Transfer the volume to a 100 mL volumetric flask. 3. Bring to volume with synthetic seawater. 4. Mix well.
Distilled water	100 mL	
General Details		
Storage	Glass or plastic container at 4°C	
Stability	One year.	
Purpose	For working standards (Low Range) preparation. Necessary if NOx only analysis will be performed, otherwise use Stock standard A to prepare combined stock standard.	

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8.5.11 Working standards

Prepare fresh working standards before each analysis run.

8.5.11.1 Combined Stock standard [5 mg/L] - This is to be prepared if running both NO_x and PO₄ analyses simultaneously (both instrument channels are used).

8.5.11.1.1 Pipette 500 µL (0.5 mL) of Nitrate Stock Standard A (1000 mg/L) into a 100 mL volumetric cylinder.

8.5.11.1.2 Pipette 500 µL (0.5 mL) of Phosphate Stock Standard A (1000 mg/L) into the same 100 mL volumetric cylinder.

8.5.11.1.3 Bring the volumetric flasks to volume with **synthetic seawater** and mix thoroughly.

8.5.11.1.4 Prepare calibration standards in individual volumetric flasks using **synthetic seawater** and the following table:

Combined Stock (mL)	Combined Stock (µL)	Combined Calibrants [mg/L]
4.0	4,000	0.200
2.0	2,000	0.100
1.0	1,000	0.050
0.5	500	0.025
0.2	200	0.010
0.1	100	0.005

8.5.11.2 NO_x Only Channel Run

8.5.11.2.1 Pipette standard C into 100 mL volumetric flasks, following the table below.

8.5.11.2.2 Dilute to volume with **synthetic seawater** and mix thoroughly.

mL of Stock Standard C (Low Range)	µL of Stock Standard C (Low Range)	Calibrants [mg/L]
1.00	1,000	0.056
0.75	750	0.042
0.50	500	0.028
0.25	250	0.014
0.10	100	0.006

9. Procedure

9.1. Sample Collection

9.1.1. Prepare a Van Dorn bottle to be deployed.

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- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close the Van Dorn bottle at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7. Dispense a few mL of the seawater into a clean 125 mL polypropylene plastic bottle to rinse the bottle 3 times.
- 9.1.8. Fill the 125 mL polypropylene plastic bottle with seawater.
- 9.1.9. Place the sample bottle inside the cooler until analysis in the lab.
 - 9.1.9.1. Samples can be preserved for 28 days at 4°C prior to analysis.
- 9.1.10. **Poison the sample (pH≈ 2)** upon arrival in the lab.
 - 9.1.10.1. Carefully add **400 µL** of 5.4 N sulfuric acid (H₂SO₄) to a 100 mL seawater sample bottle.
 - 9.1.10.2. Close the Falcon tube and invert to mix.
- 9.1.11. Analyze samples within 28 days.
 - 9.1.11.1. **Prior to analysis, the sample should be brought to neutral pH** by adding **500 µL** of 5.4 N NaOH solution to 100 ml bottles. (A random pH check after neutralization is suggested to make sure the NaOH solution was sufficiently strong to neutralize the samples.)

9.2 Sample/Instrument Pre-Run Preparation: Analysis Setup

- 9.2.1 Prepare Sampler tubes according to the Analysis Protocol with calibration standards and previously filtered samples.
 - 9.2.1.1 Pour the highest calibration standard into a 50 mL tube and position it into the calibrant rack to the left-hand side (Numbers 921 or 922).
 - 9.2.1.2 Pour the remaining calibration standards into the 10 mL small sampler cups and position them into the calibrant rack to the left-hand side (Numbers 901-920).
 - 9.2.1.3 **Filter samples to be analyzed** directly into 10 mL sampler cups and position them into the sample racks on the right-hand side (Numbers 1-120).
 - 9.2.1.3.1 Note: Ideally, one clean filter will be used for each sample, but filters may be reused with 5 mL rinses of sample water before saving the filtrate for analysis.
- 9.2.2 Create a New Run or Copy Run
 - 9.2.2.1 Click "Set Up"
 - 9.2.2.2 Click "Analysis/Run"
 - 9.2.2.3 Double-click on the "NO_x_PO₄" folder from the left side of the pop-up window.
 - 9.2.2.4 Select the NO_x_PO₄.anl run from the list in the center of the window.

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- 9.2.2.5 Click “New Run” or “Copy Run” to create the run based on the previously selected run template.
 - 9.2.2.5.1 Insert Run Name: “Date_Quarter#_NPRun#” in the provided space at the upper right-hand corner of the Main tab. (E.g. 231025_Q2_NP1)
 - 9.2.2.5.2 Check Channel tab
 - 9.2.2.5.2.1 Calibrants concentrations and units (mg/L).
 - 9.2.2.5.2.2 Spike Recovery Standard Concentration [5 mg/L]
 - 9.2.2.5.3 Modify the Tray Protocol tab to match your run according to the following criteria:
 - 9.2.2.5.3.1 1 Primer (P) and 1 Drift (D) with the highest calibrant.
 - 9.2.2.5.3.2 Calibrants + Blank (C) in decreasing order.
 - 9.2.2.5.3.3 Baseline (B), whose cup number is 0.
 - 9.2.2.5.3.4 3 Carryovers; 1 H1 using the highest calibrant and 2 L1 using the lowest calibrant.
 - 9.2.2.5.3.5 1 Recovery - Spiked Blank (K) - See Section 13 for Spike preparation instructions.
 - 9.2.2.5.3.6 1 CRM (NO₃) as QC1 - See Section 13 for CRM preparation instructions.
 - 9.2.2.5.3.7 10 Samples
 - 9.2.2.5.3.8 1 Spiked Sample (K) - See Section 13 for Spike preparation instructions.
 - 9.2.2.5.3.9 1 CRM as QC1 (NO₃)
 - 9.2.2.5.3.10 Baseline (B), whose cup number is 0.
 - 9.2.2.5.3.11 Repeat 10 samples, 1 QC, and 1 Baseline until completing the sample analysis for the day.
 - 9.2.2.5.3.12 Ensure that the tray protocol ends with 1 Drift (D) using the highest calibrant, and 1 End (E), which is the cup number 0.
 - 9.2.2.5.4 Create a New run based on a previously existing Tray Protocol:
 - 9.2.2.5.4.1 Import a previously created tray protocol from Excel. Make sure the tray protocol follows the correct format to be imported.
 - 9.2.2.5.4.2 Excel file must be closed prior to importing.
 - 9.2.2.5.5 Click “OK” to save the new run.

9.3 Autoanalyzer Instrument Start-Up

- 9.3.1 Check the LED lamp, and change it if necessary.
 - 9.3.1.1 Note: The instrument needs to be **OFF** in order to perform this step.
 - 9.3.1.2 Remove the photometer cover of Channel 1.
 - 9.3.1.3 Ensure the lamp wavelength corresponds to NO_x analysis.
 - 9.3.1.3.1 NO₃⁻/NO₂⁻ lamp: **540 nm**.

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- 9.3.1.3.2 If not, proceed to change the lamp following the instructions below.
- 9.3.1.4 Unscrew the light power cable.
- 9.3.1.5 Loosen the two screws that hold the LED lamp.
- 9.3.1.6 Replace the LED lamp.
- 9.3.1.7 Ensure the lamp is correctly placed over the holder and fasten the screws.
- 9.3.1.8 Carefully insert and tighten the light power cable.
- 9.3.1.9 Close the photometer cover.
- 9.3.2 Physically check the cadmium column in Channel 1 (NO_x).
 - 9.3.2.1 Ensure the column ends remain properly connected and there are no spills of the buffer solution on the instrument.
 - 9.3.2.2 Change the column, if necessary.
 - 9.3.2.2.1 See column preparation instructions in Section 9.7.
 - 9.3.2.2.2 Ensure no air bubbles enter the column during the mounting process by running the buffer solution (NH₄Cl) through the line (4-way valve open) and then installing the column.
 - 9.3.2.2.3 Refer to Section 9.8 to **condition the newly mounted column**.
 - 9.3.2.2.3.1 This step is performed after step 9.3.17.11, before setting the Gain.
- 9.3.3 Ensure you have a sufficient supply of reagents to complete the run.
- 9.3.4 Ensure the Channel 1 lines are correctly connected to the pump module. Lines are labeled close to the pump module, and numbers may be found engraved on a silver plate on the right side of the pump module, prior to entering the system.
 - 9.3.4.1 **Line A** to pump line #6
 - 9.3.4.2 **Line B** to pump line #8
 - 9.3.4.3 **Line C** to pump line #10
- 9.3.5 Turn ON the instrument in the following order.
 - 9.3.5.1 Computer
 - 9.3.5.2 Sampler
 - 9.3.5.3 Power Supply
- 9.3.6 Open the AACE Software.
 - 9.3.6.1 Icon at Desktop.
- 9.3.7 Login - No password
 - 9.3.7.1 Click "OK".
- 9.3.8 Confirm the analyzer and software are communicating by observing a permanent blue light in the photometer module, meaning it is ready for operation.
 - 9.3.8.1 Blinking blue light means the system is operating (charting & running).
 - 9.3.8.2 Red - STOP! Leak detected.
- 9.3.9 Place all lines inside the distilled water bottle.
 - 9.3.9.1 Ammonium Chloride line - clear line labeled "NH₄Cl"
 - 9.3.9.2 Color reagent line - blue line labeled "NO_x Color"
 - 9.3.9.3 Wash line - clear thick line labeled "Channel 1 Wash"

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- 9.3.9.4 Clear line - labeled "DI water/Channel 1"
- 9.3.9.5 Red line labeled "TKN Salicylate/NOx DI"
- 9.3.10 Place the sampler wash line into the distilled water bottle.
 - 9.3.10.1 This line is the only one on the right side of the AS2 sampler (close to the computer station).
- 9.3.11 Click "Charting" on the system window toolbar.
 - 9.3.11.1 Or click on "Run" in the main menu and select "Chart"
 - 9.3.11.2 Click "OK" if the NOx_PO4 analysis is displayed or "Change" to select NOx_PO4 analysis.
- 9.3.12 Click on the "Reagent Valve Set 1" icon on the system's main window.
 - 9.3.12.1 Select "Ch1_Ch3.rva".
 - 9.3.12.2 Click "OK".
- 9.3.13 Click on the "Pump" icon on the system's main window.
 - 9.3.13.1 Select "Normal" to pump at normal speed.
 - 9.3.13.2 Select "Engage Platen".
 - 9.3.13.2.1 Observe the platen slowly moving down. When in place, the Platen LED light on the control switch will turn green.
 - 9.3.13.3 Press "Close" to close the window.
 - 9.3.13.3.1 The pump will continue running at the selected speed.
- 9.3.14 Set Light Power.

Perform this adjustment every time an analysis is run for the first time, when pump tubes get changed, and occasionally to compensate for lamp aging.

 - 9.3.14.1 Right-click on the charting window.
 - 9.3.14.2 Select "Set Light Power" from the pop-up menu.
- 9.3.15 Ensure the Gain is set to 10.
 - 9.3.15.1 Gain value can be seen at the bottom of the charting window.
 - 9.3.15.2 If not set to 10, change the value by clicking on "Channel 1" on the system's main window and manually inserting the right value in the provided space for the Gain.
- 9.3.16 Perform Water Baseline Check.
 - 9.3.16.1 Pump distilled water through the system for 10-15 minutes at normal speed.
 - 9.3.16.1.1 Monitor the flowchart and wait for the system to stabilize.
 - 9.3.16.1.2 Visually inspect the bubble pattern through the whole system for the same size, shape, and distance. See AA500 Complete page 6.6 for bubble visual reference.
 - 9.3.16.2 **Set the base** once the system is stable.
 - 9.3.16.2.1 Right-click on the chart.
 - 9.3.16.2.2 Select "Set Base/Offset".
 - 9.3.16.2.3 Refer to "Wash Solution Baseline" in the "How To" Section if baseline does not stabilize.
 - 9.3.16.3 Annotate the water baseline value (%) in the chart's upper right corner.
 - 9.3.16.3.1 Expected value: 5.0 %.

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- 9.3.17 Perform Reagent Check.
 - 9.3.17.1 Remove the NH_4Cl and NO_x Color reagent lines from the distilled water bottle.
 - 9.3.17.2 Pat dry the reagent line tubes with a Kimwipe.
 - 9.3.17.3 Insert reagent lines into corresponding reagent solution containers.
 - 9.3.17.3.1 " NH_4Cl " into the Ammonium Chloride bottle.
 - 9.3.17.3.2 " NO_x Color" into the Color Reagent bottle.
 - 9.3.17.3.3 Refer to the system diagrams in case of doubt.
 - 9.3.17.4 Leave the remaining lines in the distilled water bottle.
 - 9.3.17.5 Cover all reagent bottles with parafilm.
 - 9.3.17.6 Continue pumping the solutions through reagent lines at normal speed for 10-15 mins.
 - 9.3.17.6.1 Visually inspect the bubble pattern and allow it to stabilize.
 - 9.3.17.7 **Switch the 4-way valve** to insert the cadmium column into the liquid stream.
 - 9.3.17.7.1 On the system's main window, double-click on the "Tray Valve" icon.
 - 9.3.17.7.2 Select "ON" to open the valve.
 - 9.3.17.7.3 The icon color will change to green, and the tray valve on the manifold will light up red.
 - 9.3.17.7.4 Click on "Close".
 - 9.3.17.8 Wait several minutes until the baseline slightly rises and ensure the baseline is stable.
 - 9.3.17.9 Annotate the reagent baseline value (%) on the chart window's upper right corner.
 - 9.3.17.10 Calculate the reagent absorbance.
 - 9.3.17.10.1 Insert the reagent baseline value (%) and the gain (10) into the corresponding space in the "Absorbance_Calcs." worksheet located in the following path: Desktop>AA500 Docs&Manuals>DailyUse
 - 9.3.17.10.2 If the resulting absorbance is >20%, reagents, and/or the wash pot may be contaminated.
 - 9.3.17.11 **Reset the Base.**
 - 9.3.17.11.1 Right-click on the chart screen.
 - 9.3.17.11.2 Select "Set Base/Offset".
 - 9.3.17.11.3 In case a new Cd column was mounted, proceed to condition the column now.
 - 9.3.17.12 Wait several minutes to ensure the baseline is stable.
- 9.3.18 Perform Sensitivity (Gain) Check.
 - 9.3.18.1 Sample the highest calibrant
 - 9.3.18.1.1 Double-click on the "Sampler" icon.
 - 9.3.18.1.2 Enter the cup number containing the highest calibrant.
 - 9.3.18.1.3 Click on "Sample".

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- 9.3.18.1.4 Pump for 2 minutes.
- 9.3.18.1.5 After 2 minutes, select “Wash”.
- 9.3.18.1.6 Close the window by clicking “Cancel”.
- 9.3.18.2 **Observe the Primer peak (1st peak) until it reaches a plateau.**
 - 9.3.18.2.1 This takes approximately 8 minutes.
 - 9.3.18.2.2 Annotate the peak value %.
- 9.3.18.3 Right-click on the chart screen and select “**Set Gain**” while the peak is still at the plateau.
- 9.3.18.4 Calculate the Sensitivity
 - 9.3.18.4.1 Complete the Sensitivity table in the “Absorbance_Calcs.” worksheet located in the following path: Desktop>AA500 Docs&Manuals>DailyUse
 - 9.3.18.4.2 The peak % used is the one observed after setting the gain (~90%)
- 9.3.18.5 **Wait until the peak fully passes and reset the base.**
 - 9.3.18.5.1 Right-click on the chart screen and select “**Set Base/Offset**”.
- 9.3.18.6 Verify the baseline and bubble pattern stability.
- 9.3.19 Proceed to analyze samples.

9.4 Start the Run

- 9.4.1 **Verify that all sample IDs refer to the correct sample rack position in the Tray Protocol before proceeding.**
- 9.4.2 Click on “Run” in the main menu.
- 9.4.3 Click on “Start”.
- 9.4.4 Double-click on the “Data” folder under the “Analysis” area.
- 9.4.5 Double-click on “NOx-PO4”.
- 9.4.6 Select the previously created .run file from the list in the center of the window.
- 9.4.7 Click “OK”.
- 9.4.8 Insert the analyst's name and comments in the provided spaces on the pop-up window.
- 9.4.9 Click “OK” to start the run.
 - 9.4.9.1 The sample protocol will start running and will stop once the last command on the Tray Protocol has been run.
 - 9.4.9.2 Wait for a pop-up window with a run completion message in the top left corner of the computer window.
 - 9.4.9.3 Continue with the Daily Shutdown instructions in Section 9.5.

9.5 Daily Shutdown

Perform this Daily shutdown if the instrument will be used for a few consecutive days; otherwise, complete a Weekly Cleanup (Refer to Section 11 for instructions.)

In any case, the **4-way valve must be closed prior to starting any cleanup process.**

- 9.5.1 Restart Charting.

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- 9.5.1.1 Once the system ends a run, it closes the charting windows and disables some software functions.
- 9.5.2 Switch the 4-way valve to remove the cadmium column from the liquid stream.
 - 9.5.2.1 On the system's main window, double-click on the "Tray valve" icon.
 - 9.5.2.2 Select "Off" to close the valve.
 - 9.5.2.3 Ensure the "Tray valve" icon color turns yellow and that the red light on top of the valve turns off.
- 9.5.3 Leave the Sampler wash line in the distilled water bottle.
- 9.5.4 Remove the reagent lines from the reagent bottles.
- 9.5.5 Pat dry the lines with a Kimwipe.
- 9.5.6 Place the reagent lines into the distilled water bottle.
- 9.5.7 Pump for a minimum of 20 minutes.
- 9.5.8 After 20 minutes, remove the lines from the distilled water bottle.
 - 9.5.8.1 Drain the system by continuing to pump. Observe the lines in the chemistry module until most bubbles have exited the system.
- 9.5.9 Pat dry the tubes and store them dry inside the provided flasks.
- 9.5.10 Turn off the pump and release the platen.
 - 9.5.10.1 Click on the "Pump" icon on the main window.
 - 9.5.10.2 Switch the pump off.
 - 9.5.10.3 Click on "Release platen".
- 9.5.11 Stop Charting.
 - 9.5.11.1 Click the "Stop" button in the System window toolbar.
 - 9.5.11.1.1 Or click the "Close" button in the charting window.
- 9.5.12 Turn OFF the instrument in the following order.
 - 9.5.12.1 Power Supply
 - 9.5.12.2 Sampler
 - 9.5.12.3 Computer
- 9.5.13 Discard samples and calibrants from the Sampler cups in the provided waste container.
- 9.5.14 Wash sample cups.
 - 9.5.14.1 Rinse cups with tap water 3 times.
 - 9.5.14.2 Submerge the cups in a 10% HCl bath and let soak overnight.
 - 9.5.14.3 After use, properly store any remaining HCl to be reused several times.
 - 9.5.14.3.1 Discard into the corresponding unwanted material container if turbid.
 - 9.5.14.4 Triple-rinse the sample cups with distilled water.
 - 9.5.14.5 Let the cups dry standing on a dish rack or inside the oven.

9.6 Data Retrieval

- 9.6.1 Reports
 - 9.6.1.1 Click on "Retrieve" on the main menu.
 - 9.6.1.2 Select "Report" from the dropdown.
 - 9.6.1.3 Double-click on the desired analysis (menu on the left window)

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- 9.6.1.4 Double-click on the desired .run File.
- 9.6.1.5 Select the desired option from the buttons at the bottom of the report window.
 - 9.6.1.5.1 You may save a PDF version of the report at this point.
- 9.6.2 ASCII files
 - 9.6.2.1 Click on “Retrieve” on the main menu.
 - 9.6.2.2 Select “View Chart” from the dropdown.
 - 9.6.2.3 Double-click on the desired analysis (menu on the left window).
 - 9.6.2.4 Double-click on the desired .run File.
 - 9.6.2.5 On the icons on top of the chart window, click on the “Export to ASCII file” icon (3rd left to right)
 - 9.6.2.6 Format Tab
 - 9.6.2.6.1 Format should always be set to “Text Format (ASCII)”
 - 9.6.2.6.2 Confirm the Directory to be used
 - 9.6.2.7 Export Data Tab
 - 9.6.2.7.1 Check all desired information to be included in the file
 - 9.6.2.7.2 Go to the “Preview Tab” to confirm the selection
 - 9.6.2.8 Click on “Export” to save the file.

9.7 Cadmium Column Preparation

- 9.7.1 Refer to AA500 Complete User Manual (pp. 6.13-6.15/PDF page 69 - 71) or to AA500 Method No. A002-21 (p. 7) (User manual handout at the lab) for detailed instructions on the following procedures:
 - 9.7.1.1 Preparing cadmium particles.
 - 9.7.1.2 Packing the column.
 - 9.7.1.3 Cadmium column conditioning.
- 9.7.2 Perform this process under a hood and follow all safety precautions.
- 9.7.3 If more than one column is prepared, additional spare columns can be stored indefinitely, sealed in a diluted buffer solution (Ammonium chloride buffer solution).
- 9.7.4 The Cadmium column may be kept until the reduction efficiency reaches unacceptable levels.

9.8 Cadmium Column Conditioning

The conditioning process is done **only** the first time running a new column.

- 9.8.1 Switch the 4-way valve to insert the cadmium column into the liquid stream.
 - 9.8.1.1 On the system’s main window, double-click on the “Tray Valve” icon.
 - 9.8.1.2 Select “Dilute/ON” to open the valve.
 - 9.8.1.3 The icon color will change to green.
- 9.8.2 Place 2 mg N/L **Nitrate** standard in a sampler cup and sample for **5 minutes**.
- 9.8.3 Place 2 mg N/L **Nitrite** standard in a sampler cup and sample for **10 minutes**.
- 9.8.4 Begin sample analysis following instructions in Section 9.4.

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9.8.5 Old cadmium may be reused. If discarded, handle and dispose of it as hazardous waste.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Field forms and Maintenance sheets shall be kept in the designated file cabinet at the CCRI office.
- 10.3. Daily/Weekly/Monthly/Yearly maintenance shall be recorded in maintenance sheets.
- 10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.5. All spreadsheets should be uploaded to the internal CCRI Water Quality Database.
- 10.6. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. The autoanalyzer exterior may be cleaned using an alcohol-damped, lint-free cloth.
- 11.2. Refer to “AA500 Complete User Manual” or “AA500 Operation Manual” for specific instrument maintenance instructions.
- 11.3. Recommended pump tubes use is for a maximum of 200 hours.
- 11.4. Autoanalyzer Daily Cleanup
 - 11.4.1. Follow Daily Shutdown instructions described in Section 9.5.
 - 11.4.2. Document your cleanup method in the provided lab notebook.
- 11.5. Autoanalyzer Weekly Cleanup
 - 11.5.1. Select “All Wash” Mode by pressing the reagent valve set 1 icon
 - 11.5.2. Pump the following on fast mode through all lines for at least 2 minutes each.
 - 11.5.2.1. NaOH (Sodium hydroxide)
 - 11.5.2.2. Distilled water
 - 11.5.2.3. HCl 1N (Hydrochloric acid)
 - 11.5.2.4. Distilled water
 - 11.5.2.5. Special Wash Solution 10% (Sodium hypochlorite)
 - 11.5.2.6. Distilled water
 - 11.5.3. Remove the lines from the Distilled water bottle and store them inside their respective channel storage flask.
 - 11.5.4. Leave the pump running for a few minutes to drain the system.
 - 11.5.4.1. Observe until no liquid is seen inside the system and/or the waste line.
 - 11.5.5. Once the system has drained, stop Charting.
 - 11.5.5.1. Click the “Stop” button in the System window toolbar.
 - 11.5.5.1.1. Or click the “Close” button in the charting window.
 - 11.5.6. Print and fill out the Weekly Maintenance Sheet included in the AA500 Complete User Manual.
 - 11.5.6.1. Save the sheet in the instrument’s maintenance binder.
- 11.6. Autoanalyzer Monthly Cleanup
 - 11.6.1. Disconnect the cadmium column from the flow

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- 11.6.2. Wash the manifold and sample line with the special wash solution (diluted hypochlorite solution) for 10 minutes.
- 11.6.3. Follow the previous wash with distilled water flushing all reagent lines for 10 minutes.
- 11.6.4. Print and fill out the Monthly Maintenance Sheet included in the user manual.
 - 11.6.4.1. Save the sheet in the instrument's maintenance binder.
- 11.7. For Monthly and Yearly Maintenance follow the instructions provided in the User Manuals and complete the corresponding documentation.
 - 11.7.1. AA500 Operation Manual - Section 5
 - 11.7.2. AA500 Complete User Manual - Section 4

12. Troubleshooting

- 12.1 Refer to AA500 Complete User Manual (Section 8) For Troubleshooting instructions.
- 12.2 Refer to AA500 Complete User Manual (Section 6.10) for sensitivity and noise troubleshooting instructions.

13. QA/QC

- 13.1 As per QAPP Table 4, seawater $\text{NO}_3^-/\text{NO}_2^-$ should be measured in mg/L with an accuracy of 0.004 mg/L ($\pm 20\%$) and a precision of $<20\%$.
- 13.2 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.
- 13.3 Individual CRM Preparation
 - 13.3.1 Make CRM dilution fresh before each analysis.
 - 13.3.2 Nitrate - Pipette 1,200 μL (1.2 mL) of Nitrate CRM [2.5 mg/L] solution and bring to volume with synthetic seawater in a 100 mL volumetric flask. This dilution will have a concentration of 0.030 mg/L.
- 13.4 Combined ERA CRM Preparation
 - 13.4.1 Initial Dilution as instructed by the manufacturer- Pipette 2,500 μL (2.5 mL) of concentrated ERA CRM vial and bring to volume with **synthetic seawater** in a 500 mL volumetric flask. This dilution will have a concentration of 23 mg/L $\text{NO}_3\text{-N}$. (Usually good for up to 6 months when stored cold)
 - 13.4.2 Pipette 650 μL (0.65 mL) of previously diluted ERA CRM [23 mg/L $\text{NO}_3\text{-N}$] and bring to volume with **synthetic seawater** in a 100 mL volumetric flask. This dilution will have a concentration of **0.15 mg/L**.
- 13.5 Spike Preparation (total vol. = 5 mL, ~2 runs)
 - 13.5.1 Pipette 4,850 μL of Synthetic Sea Water/Sample into a sample cup
 - 13.5.2 Add 150 μL (0.15 mL) of Combo Stock Solution [5 mg/L] into the sample cup
 - 13.5.3 Cap the tube and gently invert it to mix the contents.
 - 13.5.4 This will add **0.15 mg/L** NO_x to the sample.
- 13.6 Spike Preparation (total vol. = 10 mL)
 - 13.6.1 Pipette 9,970 μL of Synthetic Sea Water/Sample into a sample cup
 - 13.6.2 Add 300 μL (0.30 mL) of Combo Stock Solution [5 mg/L] into the sample cup

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13.6.3 Cap the tube and gently invert it to mix the contents.

13.6.4 This will add 0.15 mg/L NO_x to the sample.

14. References

- 14.1. U.S. ENVIRONMENTAL PROTECTION AGENCY. 1993. Methods for Determination of Inorganic Substances in Environmental Samples; EPA-600/R/93/100—Draft. Environmental Monitoring Systems Lab., Cincinnati, Ohio.
- 14.2. AA500 Complete User Manual.
- 14.3. AA500 Operational Manual.
- 14.4. AACE Software Manual.